Module 7: Molecular Genetics — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. State the Central Dogma of Molecular Biology: DNA → RNA → Protein
2. Describe the structure of DNA (double helix, nucleotides, base pairing rules)
3. Explain the process of DNA replication and why it is semi-conservative
4. Describe transcription: how DNA is copied into mRNA
5. Describe translation: how mRNA is decoded into a polypeptide at the ribosome
6. Explain the roles of mRNA, tRNA, and rRNA in protein synthesis
7. Use the genetic code (codon table) to determine amino acid sequences

The Central Dogma

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  DNA  →  RNA  →  Protein
       ↑          ↑
Replication  Transcription  Translation
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- **DNA** stores genetic information (the "blueprint")
- **RNA** carries and translates the information (the "messenger" and "decoder")
- **Protein** performs cellular functions (the "worker")

Key Terms to Know

DNA Structure

- **Nucleotide** — Monomer of nucleic acids: phosphate group + sugar + nitrogenous base
- **Deoxyribose** — The sugar in DNA nucleotides
- **Nitrogenous Bases** — Adenine (A), Thymine (T), Guanine (G), Cytosine (C)

- **Base Pairing Rules** — A pairs with T (2 hydrogen bonds); G pairs with C (3 hydrogen bonds)
- **Double Helix** — Two antiparallel strands wound around each other
- **Antiparallel** — The two DNA strands run in opposite directions (5′ → 3′ and 3′ → 5′)
- **Complementary** — Each strand determines the sequence of the other via base pairing
- **Chromosome** — A long, continuous DNA molecule packaged with histone proteins
- **Gene** — A segment of DNA that encodes a functional product (usually a protein)

DNA Replication

- **Semi-Conservative Replication** — Each new DNA molecule has one original strand and one new strand
- **Helicase** — Enzyme that unwinds and separates the double helix
- **DNA Polymerase** — Enzyme that synthesizes new DNA by adding nucleotides (5′ → 3′ direction only)
- **Leading Strand** — Synthesized continuously in the direction of the replication fork
- **Lagging Strand** — Synthesized in short fragments (Okazaki fragments) away from the fork
- **Primase** — Enzyme that makes a short RNA primer to start replication
- **Ligase** — Enzyme that joins Okazaki fragments together
- **Origin of Replication** — Where replication begins; produces a replication bubble

Transcription (DNA → RNA)

- **RNA** — Single-stranded nucleic acid; uses ribose sugar and uracil (U) instead of thymine (T)
- **mRNA (Messenger RNA)** — Carries the genetic message from DNA to the ribosome
- **RNA Polymerase** — Enzyme that synthesizes mRNA using the DNA template strand
- **Template Strand** — The DNA strand read by RNA polymerase (3′ → 5′)
- **Coding Strand** — The DNA strand NOT read; has same sequence as mRNA (except T→U)
- **Promoter** — DNA sequence that signals where transcription begins
- **Terminator** — DNA sequence that signals where transcription ends

- **5′ Cap and Poly-A Tail** — Modifications added to eukaryotic mRNA for protection and export
- **Introns** — Non-coding sequences removed during RNA processing (spliced out)
- **Exons** — Coding sequences retained in the mature mRNA

Translation (RNA → Protein)

- **Codon** — A three-nucleotide sequence on mRNA that specifies one amino acid
- **Genetic Code** — The set of rules mapping codons to amino acids (64 codons, 20 amino acids)
- **Start Codon** — AUG (codes for methionine; signals where translation begins)
- **Stop Codons** — UAA, UAG, UGA (signal end of translation; no amino acid)
- **tRNA (Transfer RNA)** — Carries amino acids to the ribosome; has an anticodon that pairs with mRNA codon
- **Anticodon** — Three-nucleotide sequence on tRNA complementary to the mRNA codon
- **Ribosome** — The molecular machine that reads mRNA and assembles the polypeptide
- **rRNA (Ribosomal RNA)** — Structural and catalytic component of ribosomes
- **A Site (Aminoacyl)** — Ribosome site where incoming charged tRNA binds
- **P Site (Peptidyl)** — Ribosome site where the growing polypeptide chain is held
- **E Site (Exit)** — Ribosome site where empty tRNA exits
- **Polypeptide** — A chain of amino acids linked by peptide bonds; folds into a functional protein

DNA Replication: Step by Step

1. **Helicase** unwinds the double helix at the origin of replication
2. **Primase** adds a short RNA primer
3. **DNA Polymerase** adds complementary nucleotides (A-T, G-C) in the 5′ → 3′ direction
4. **Leading strand** is synthesized continuously; **lagging strand** in Okazaki fragments
5. **Ligase** seals the fragments together
6. Result: **Two identical DNA molecules**, each with one old and one new strand (semi-conservative)

Transcription: Step by Step

1. **RNA Polymerase** binds to the **promoter** on DNA
2. RNA Polymerase reads the **template strand** (3′ → 5′) and builds mRNA (5′ → 3′)
3. Base pairing: A→U, T→A, G→C, C→G (note: U replaces T in RNA)
4. RNA Polymerase reaches the **terminator** → mRNA is released
5. In eukaryotes: **5′ cap** and **poly-A tail** are added; **introns are spliced out**
6. Mature mRNA exits the nucleus through nuclear pores → heads to the ribosome

Translation: Step by Step

1. **mRNA** binds to the ribosome; the start codon (**AUG**) is positioned at the P site
2. A **tRNA** with the anticodon matching the A site codon brings the next amino acid
3. A **peptide bond** forms between the amino acids (catalyzed by rRNA)
4. The ribosome shifts one codon forward (**translocation**): tRNA moves P→E (exits), new tRNA enters A site
5. Steps 2-4 repeat until a **stop codon** (UAA, UAG, or UGA) is reached
6. The polypeptide is **released** and folds into a functional protein

Key Comparisons

Feature	DNA Replication	Transcription	Translation
Template	DNA	DNA	mRNA
Product	DNA	mRNA	Polypeptide
Enzyme	DNA Polymerase	RNA Polymerase	Ribosome
Location	Nucleus	Nucleus	Cytoplasm (ribosome)
Direction	5′ → 3′	5′ → 3′ (mRNA)	5′ → 3′ (read direction)

Study Tips

1. **Master the Central Dogma flow** — DNA → (transcription) → mRNA → (translation) → Protein
2. **Know your base pairing** — DNA: A-T, G-C. RNA: A-U, G-C
3. **Practice codon reading** — Given a DNA strand, write the mRNA, then use the codon table to find amino acids
4. **Compare the three processes** — Make a table with Location, Enzyme, Template, Product for each
5. **Remember the RNA types** — mRNA carries the message, tRNA carries amino acids, rRNA builds the ribosome
6. **Understand introns vs. exons** — "EX-ons are EX-pressed; IN-trons stay IN the nucleus"
7. **Think about why this matters** — Every trait you have comes from proteins, and every protein comes from genes via this pathway